



INTRODUCTION TO BUILDING MATERIALS - GLASS

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Born from Fusion: The Elemental Paradox

The Material

Glass is a hard substance, innately brittle, manifesting in transparent or translucent states.

The Fusion Process

It is manufactured by fusing raw sand with lime, soda, and specific admixtures, followed by rapid cooling.

The Architectural Purpose

This elemental transformation yields a material foundational to modern engineering and architectural construction.



Transparent

Translucent

Opaque

The 6 Engineering Properties of Glass



Transparency

Allows vision of the outside world.
Can be bilateral or unilateral (acting as a mirror on one side).



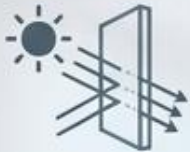
Strength

Governed by the modulus of rupture.
Inherently brittle, but engineered for immense strength via admixtures and laminations.



Workability

Highly malleable; can be molded into any shape or blown during the melting phase.



Transmittance

The specific fraction of visible light capable of passing through the material.



U-Value

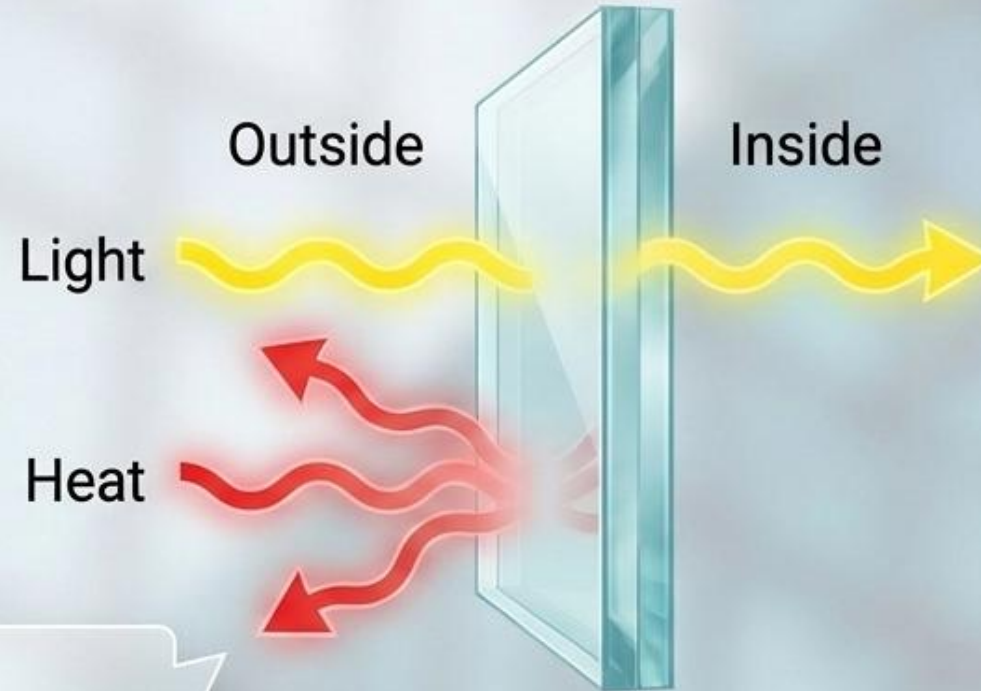
The metric of thermal transfer.
Lower U-value indicates superior insulation.



Recycle Property

100% recyclable, consistently repurposed as raw material in the construction industry.

The Physics of Comfort: Transmittance vs. U-Value



Visible Transmittance
Letting the light in. The fraction of visible light passing through the glass.

U-Value
Keeping the heat out. U-value represents the amount of heat transferred through the glass.

The Engineering Goal
An optimal insulated glass unit requires high visible transmittance (for natural lighting) combined with a highly lowered U-value (for thermal regulation).

Form Follows Function: The 3 Families of Architectural Glass

The Protectors

Engineered for structural integrity, load-bearing capability, and impact safety.

Shatterproof

Laminated

Toughened

The Regulators

Engineered to control the environment, managing thermal transfer, acoustics, and maintenance.

Insulated

Glass Wool

Glass Blocks

Extra Clean

The Chameleons

Engineered for visual adaptability, privacy control, and aesthetic variation.

Float

Chromatic

Tinted

Not All Glass Is Created Equal

Glass is manufactured, treated, and engineered differently depending on the job it needs to do — safety, insulation, privacy, or pure aesthetics. Understanding these differences helps you choose the right glass for the right application.

1

Safety & Security

Tempered, laminated, and wired glass protect people and property.

2

Energy Performance

Insulated units and Low-E coatings cut heating and cooling costs.

3

Light & Privacy

Frosted, tinted, and patterned glass balance daylight with discretion.

4

Aesthetics & Form

Mirror, reflective, and extra-clear glass shape how spaces look and feel.

1

Float Glass

The foundation of nearly every other glass type.

- ❖ Manufactured by floating molten glass over a bath of molten tin to produce a smooth and uniform surface.
- ❖ Provides excellent optical clarity with a flat, distortion-free finish.
- ❖ Available in various thicknesses and can be further processed into tempered, laminated, or insulated glass.
- ❖ Commonly used in windows, doors, mirrors, facades, furniture, and automotive applications.
- ❖ Offers good transparency, durability, and ease of fabrication for a wide range of construction uses.

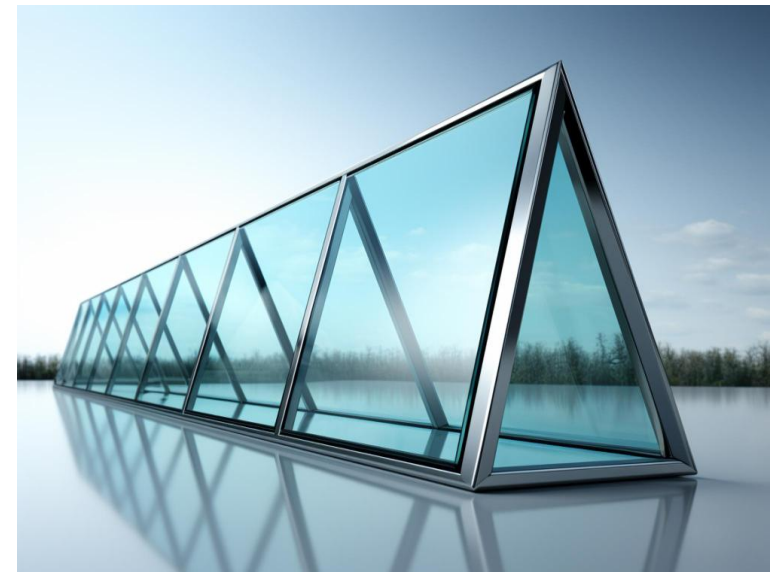
WHERE IT'S USED

Standard windows

Shopfronts & glazing

Base stock for processing

Furniture tabletops



2

Tempered Glass

Engineered to break safely, not sharply.

- ❖ Manufactured by heating ordinary glass and rapidly cooling it to increase its strength.
- ❖ About **4–5 times stronger** than ordinary (annealed) glass.
- ❖ Breaks into small, blunt fragments instead of sharp shards, reducing the risk of injury.
- ❖ Commonly used in doors, windows, shower enclosures, facades, railings, and automobile side and rear windows.
- ❖ Provides high impact resistance and thermal resistance but cannot be cut or drilled after tempering.

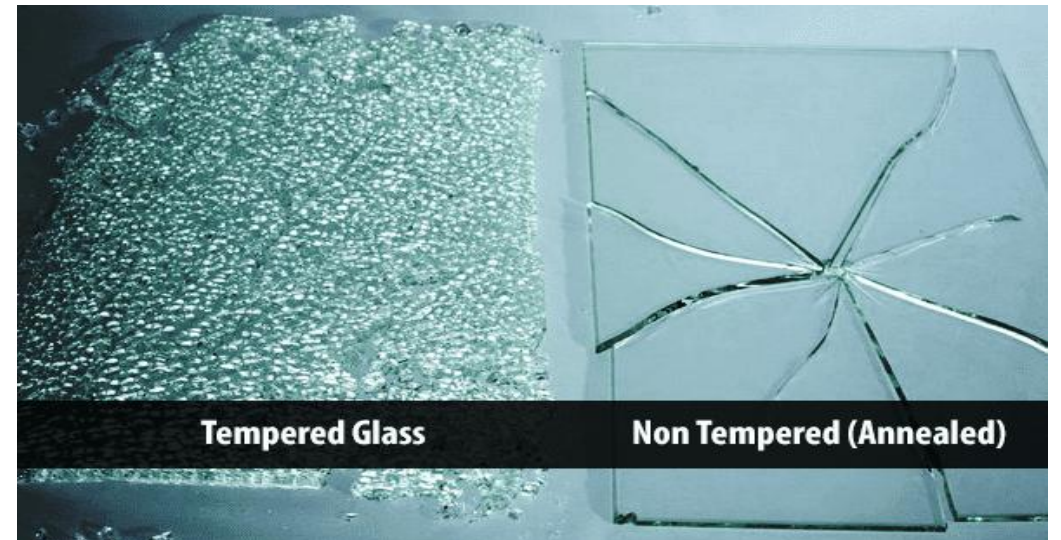
WHERE IT'S USED

Shower doors & partitions

Vehicle side windows

Glass entry doors

Sports arena backboards



3

Laminated Glass

Layered strength that holds together on impact.

- ❖ Made by bonding two or more glass sheets with a PVB or SGP interlayer.
- ❖ Provides high safety as broken glass fragments remain attached to the interlayer.
- ❖ Offers excellent impact resistance, sound insulation, and blocks up to 99% of UV rays.
- ❖ Commonly used in facades, skylights, glass railings, windows, and automobile windshields.
- ❖ More expensive and heavier than ordinary glass and requires proper edge sealing for durability.

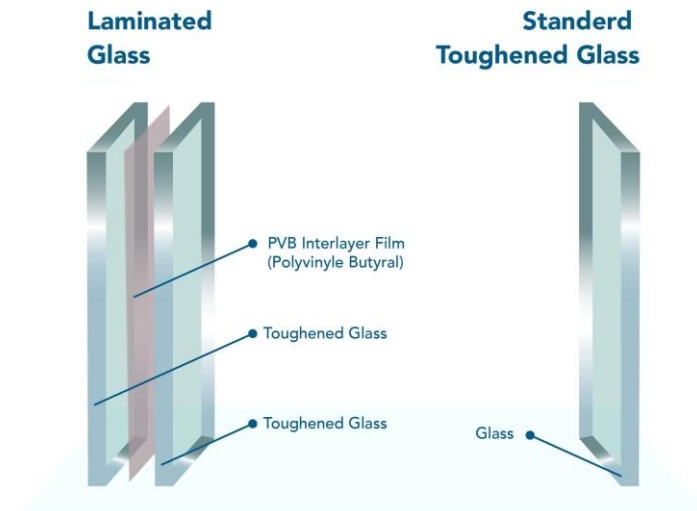
WHERE IT'S USED

Windshields

Hurricane / storm windows

Skylights & overhead glazing

Bank & security glazing



QUICK CHECK · 1

Which glass type is designed to break into small, blunt granules instead of sharp shards?

A Float Glass — the flat, clear base material

B Tempered Glass — heated and rapidly cooled for strength

✓ CORRECT

C Laminated Glass — layered with a plastic interlayer

D Wired Glass — reinforced with embedded wire mesh

Think back to how each type is manufactured before you pick.

4

Frosted Glass

Privacy without losing the light.

- ❖ Manufactured by acid etching or sandblasting one surface to create a translucent, matte finish.
- ❖ Provides privacy while allowing natural light to pass through.
- ❖ Reduces glare and diffuses light, creating a soft and uniform illumination.
- ❖ Commonly used in bathroom windows, office partitions, shower enclosures, and decorative doors.
- ❖ Requires regular cleaning as fingerprints and stains are more visible on the frosted surface.

WHERE IT'S USED

Bathroom windows

Office partitions

Shower enclosures

Decorative signage panels



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Tinted Glass

Color engineered to manage heat and glare.

- ❖ Manufactured by adding metal oxides during the glass-making process to produce colors such as bronze, grey, green, or blue.
- ❖ Reduces heat gain and glare by absorbing a portion of the sun's radiation.
- ❖ Provides privacy by limiting visibility from the outside during daylight.
- ❖ Commonly used in building facades, windows, doors, curtain walls, and automobiles.
- ❖ Helps improve energy efficiency and enhances the aesthetic appearance of buildings.

WHERE IT'S USED

Car & bus windows

Building façades

Sunroom glazing

Privacy glass panels



6

Wired Glass

A safety net built right into the pane.

- ❖ Manufactured by embedding a wire mesh within the glass during the manufacturing process.
- ❖ Holds broken glass pieces together, reducing the risk of falling shards.
- ❖ Provides improved fire resistance by delaying the spread of flames and smoke.
- ❖ Commonly used in fire doors, windows, skylights, industrial buildings, and stairwells.
- ❖ Has lower impact strength and optical clarity compared to other types of safety glass.

WHERE IT'S USED

Fire doors & fire windows

Stairwell glazing

Security-sensitive openings

Industrial skylights



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Insulated Glass Unit (IGU)

Two panes, one job: keeping temperature where it belongs.

- ❖ Consists of two or more glass panes separated by a sealed air or inert gas (e.g., argon) filled space.
- ❖ Provides excellent thermal insulation, reducing heat transfer and improving energy efficiency.
- ❖ Reduces outside noise, improving indoor acoustic comfort.
- ❖ Commonly used in energy-efficient windows, curtain walls, skylights, and commercial buildings.
- ❖ Helps prevent condensation on the inner glass surface and lowers heating and cooling costs.

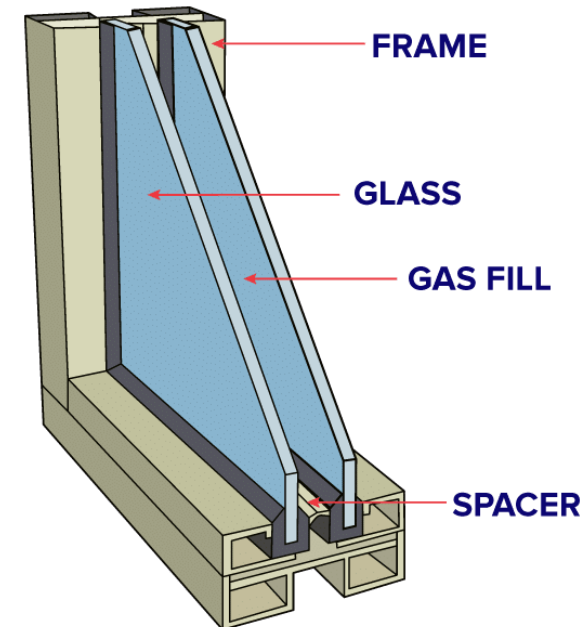
WHERE IT'S USED

Residential double-glazed windows

Cold climate construction

Curtain wall systems

Energy-efficient retrofits



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Low-E Glass

An invisible coating that does the heavy lifting.

- ❖ Coated with a thin, transparent low-emissivity (Low-E) metallic layer to reduce heat transfer.
- ❖ Reflects infrared heat while allowing natural daylight to pass through.
- ❖ Improves energy efficiency by reducing heating and cooling loads.
- ❖ Minimizes UV radiation, helping protect furniture, flooring, and fabrics from fading.
- ❖ Commonly used in energy-efficient windows, curtain walls, skylights, and green buildings.

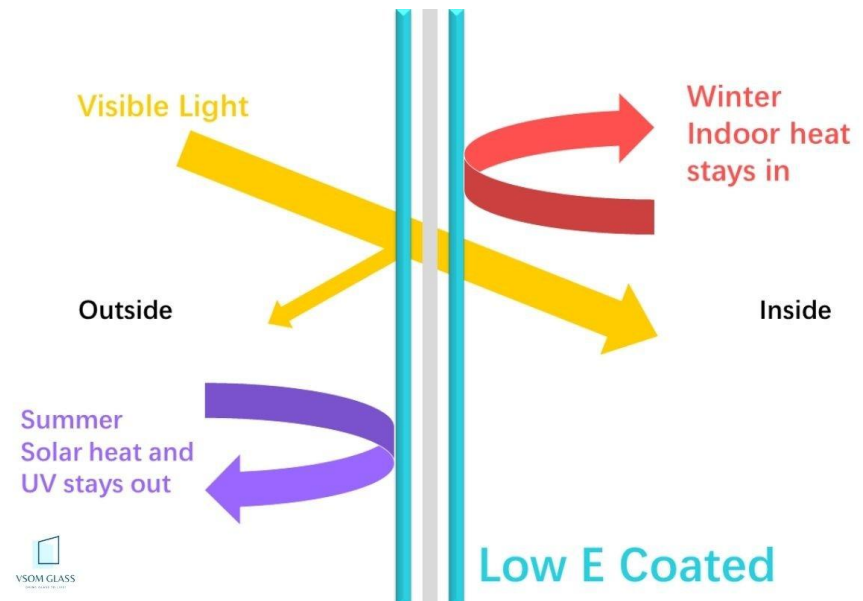
WHERE IT'S USED

Energy-efficient windows

Commercial curtain walls

Green-building certified projects

Cold-storage glazing



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Patterned / Textured Glass

Decoration and privacy pressed right into the surface.

- ❖ Manufactured by rolling or embossing patterns onto one or both glass surfaces during production.
- ❖ Provides privacy while allowing natural light to pass through.
- ❖ Available in a wide variety of decorative patterns, textures, and designs.
- ❖ Commonly used in doors, windows, partitions, shower enclosures, and interior decorative applications.
- ❖ Enhances the aesthetic appearance of buildings while reducing direct visibility.

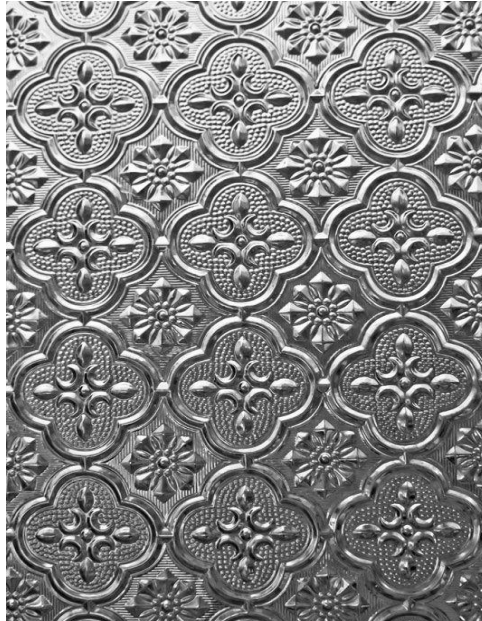
WHERE IT'S USED

Interior doors & partitions

Decorative room dividers

Cabinet glass inserts

Bathroom & shower panels



QUICK CHECK · 2

What is the main function of Low-E (Low-Emissivity) glass?

A Adding color pigment for privacy and glare control

B Reflecting infrared heat while still letting visible light through

✓ CORRECT

C Embedding wire mesh for fire-code compliance

D Removing iron content for a crystal-clear look

Hint: the 'E' stands for emissivity — think heat, not color.

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Fire-Rated Glass

Built to hold the line when it matters most.

- ❖ Specially manufactured to withstand high temperatures and resist the spread of fire and smoke.
- ❖ Maintains its integrity for a specified fire-resistance period (e.g., 30, 60, or 120 minutes).
- ❖ Provides safety by allowing visibility while acting as a fire barrier during emergencies.
- ❖ Commonly used in fire doors, stairwells, escape routes, partitions, and commercial buildings.
- ❖ Meets fire safety standards and building code requirements for fire-resistant construction.

WHERE IT'S USED

Fire doors & exit corridors

Stairwell enclosures

Compartmentalized commercial floors

Hospital & school partitions



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Glass Block

Solid, architectural, and full of character.

- ❖ Made from two molded glass halves fused together to form a hollow or solid glass block.
- ❖ Allows natural light to pass through while providing privacy.
- ❖ Offers good thermal insulation, sound insulation, and weather resistance.
- ❖ Commonly used in walls, partitions, bathrooms, stairwells, and decorative facades.
- ❖ Durable, low-maintenance, and enhances the aesthetic appearance of buildings.

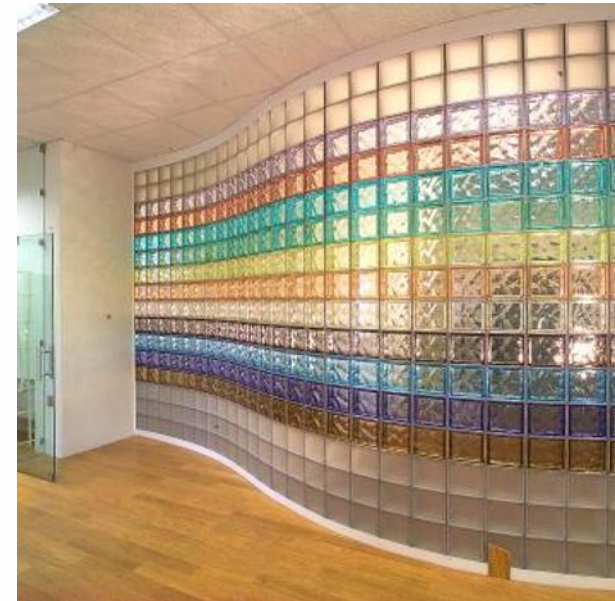
WHERE IT'S USED

Shower & bathroom walls

Basement & stairwell windows

Decorative interior walls

Security-conscious openings





REFLECTION BREAK · 2

Which glass would you specify for a high-rise office façade — and what would you trade off?

- Weigh cost, energy performance, safety, and appearance against each other.
- Would reflective glass, Low-E glass, or an insulated unit best balance those priorities?

REFLECTION BREAK — take a minute before moving on

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Glass Wool

Clarity with the green tint removed.

- ❖ Made from fine fibers of molten glass spun into a lightweight insulating material.
- ❖ Provides excellent thermal insulation, reducing heat transfer and improving energy efficiency.
- ❖ Offers good sound absorption, making it effective for acoustic insulation.
- ❖ Commonly used in roofs, walls, ceilings, HVAC ducts, and industrial equipment.
- ❖ Lightweight, non-combustible, and resistant to moisture, corrosion, and pests.

WHERE IT'S USED

HVAC Ducts

Roof Insulation

Boilers

Pipes



QUICK CHECK · 3

Which glass type is rated to withstand flames for a specific time period, such as 30, 60, or 90 minutes?

A Glass Block — thick, fused architectural units

B Reflective Glass — coated to bounce back sunlight

C Fire-Rated Glass — treated for certified fire resistance

D Mirror Glass — backed with a reflective coating

Look for the type built specifically around a time rating.

PUTTING IT ALL TOGETHER

Choosing Glass by Priority

Safety First

- Tempered Glass
- Laminated Glass
- Wired Glass
- Fire-Rated Glass

Energy Performance

- Insulated Glass (IGU)
- Low-E Glass
- Tinted Glass
- Reflective Glass

Privacy & Light Control

- Frosted Glass
- Patterned Glass
- Glass Block
- Tinted Glass

Clarity & Aesthetics

- Float Glass
- Extra-Clear Glass
- Mirror Glass
- Patterned Glass



FINAL REFLECTION

If you were specifying glass for your next project, which type would you choose — and why?

- Revisit the four priorities: safety, energy performance, privacy, and aesthetics.
- Which single glass type comes closest to balancing all four for your scenario?
- What's one piece of glass around you right now — and what type do you think it is?

REFLECTION BREAK — take a minute before moving on



Thank You